矢量分析

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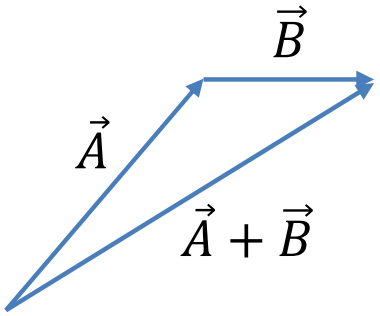
五．狄拉克δ函数

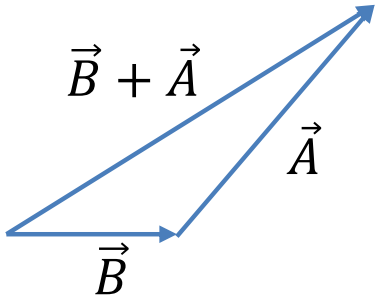
一、矢量代数

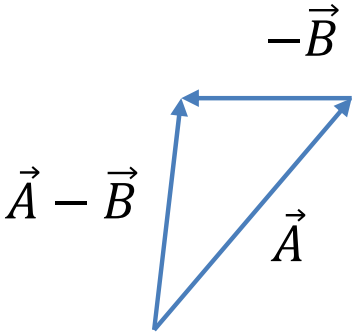
1.1 矢量运算

- 什么是矢量？既包含大小，也包含方向的量，例如：位移、速度、加速度、动量等
- 矢量运算包括四种：加法、三类乘法

（i）两个矢量相加，满足平行四边形法则，交换律





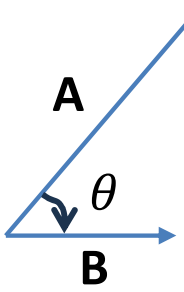


一、矢量代数

（ii）矢量乘于一个标量，大小变化，方向不变或反转，满足交换律，分配律

$a(\vec{A}+\vec{B})=a\vec{A}+a\vec{B}$

（iii）两个矢量的点乘（内积、标积）。定义 $\vec{A}\cdot\vec{B}=AB\cos(\theta)$



满足交换律、分配律 $\vec{A}\cdot\vec{B}=\vec{B}\cdot\vec{A}$

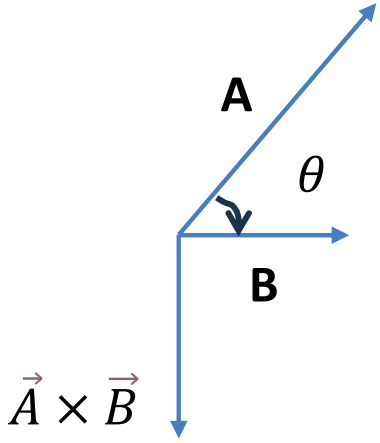
$$\vec{A}\cdot(\vec{B}+\vec{C})=\vec{A}\cdot\vec{B}+\vec{A}\cdot\vec{C}$$

（iV）两个矢量的叉乘。定义 $\vec{A}\times\vec{B}=AB\sin\theta\hat{n}$

满足分配律，不满足交换律

$$\vec{A}\times\vec{B}=-\vec{B}\times\vec{A}$$

$$\vec{A}\times(\vec{B}+\vec{C})=\vec{A}\times\vec{B}+\vec{A}\times\vec{C}$$

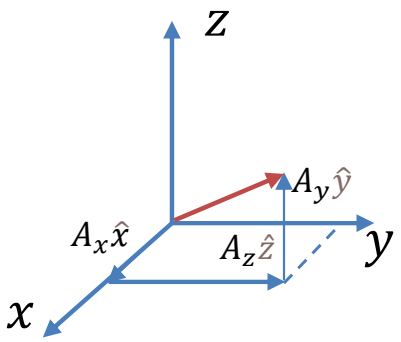


右手螺旋规则

1.2 矢量运算-分量形式

在特定的坐标系，将矢量在坐标基矢中展开，以直角坐标系为例

$$\vec{A} = A_x\hat{x} + A_y\hat{y} + A_z\hat{z}$$



(i) 矢量相加

$$\vec{A} + \vec{B} = (A_x\hat{x} + A_y\hat{y} + A_z\hat{z}) + (B_x\hat{x} + B_y\hat{y} + B_z\hat{z})$$

$$= (A_x + B_x)\,\hat{x} + (A_y + B_y)\,\hat{y} + (A_z + B_z)\,\hat{z}$$

(ii) 数乘

$$a\vec{A} = (aA_x)\hat{x} + (aA_y)\hat{y} + (aA_z)\hat{z}$$

(iii) 点乘、标量积

$$\hat{x}\cdot\hat{x} = \hat{y}\cdot\hat{y} = \hat{z}\cdot\hat{z} = 1$$

一、矢量代数

(iii) 点乘

$$\hat{x}\cdot\hat{y}=\hat{y}\cdot\hat{z}=\hat{z}\cdot\hat{x}=0$$

$$\vec{A}\cdot\vec{B}=(A_x\hat{x}+A_y\hat{y}+A_z\hat{z})\cdot(B_x\hat{x}+B_y\hat{y}+B_z\hat{z})=A_xB_x+A_yB_y+A_zB_z$$

$$\left|\vec{A}\right|=\sqrt{{A_x}^2+A_y^2+A_z^2}$$

$$\vec{A}\cdot\vec{A}={A_x}^2+A_y^2+A_z^2$$

一、矢量代数

(iv) 叉乘、矢量积

$$\hat{x}\times\hat{x}=\hat{y}\times\hat{y}=\hat{z}\times\hat{z}=0$$
$$\hat{x}\times\hat{y}=-\hat{y}\times\hat{x}=\hat{z}$$
$$\hat{y}\times\hat{z}=-\hat{z}\times\hat{y}=\hat{x}$$
$$\hat{z}\times\hat{x}=-\hat{x}\times\hat{z}=\hat{y}$$

$$\vec{A}\times\vec{B}=(A_x\hat{x}+A_y\hat{y}+A_z\hat{z})\times(B_x\hat{x}+B_y\hat{y}+B_z\hat{z})$$
$$=A_xB_y\,\hat{z}+A_xB_z(-\hat{y})+A_yB_x(-\hat{z})+A_yB_z\,\hat{x}+A_zB_x\,\hat{y}+A_zB_y(-\hat{x})$$
$$=(A_yB_z-A_zB_y)\hat{x}+(A_zB_x-A_xB_z)\hat{y}+(A_xB_y-A_yB_x)\hat{z}$$

$$\vec{A}\times\vec{B}=\begin{vmatrix}\hat{x} & \hat{y} & \hat{z}\\ A_x & A_y & A_z\\ B_x & B_y & B_z\end{vmatrix}$$

一、矢量代数

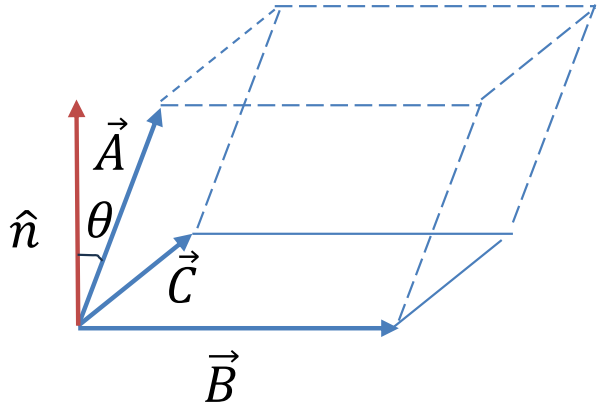
1.3 三重积

(i) 标量三重积

$\vec{A}\cdot(\vec{B}\times\vec{C})$，物理意义：其绝对值$|\vec{A}\cdot(\vec{B}\times\vec{C})|$为体积

$$\vec{A}\cdot(\vec{B}\times\vec{C})=\vec{B}\cdot(\vec{C}\times\vec{A})=\vec{C}\cdot(\vec{A}\times\vec{B})$$

$$\vec{A}\cdot(\vec{B}\times\vec{C})=\begin{vmatrix}A_x & A_y & A_y\\ B_x & B_y & B_z\\ C_x & C_y & C_z\end{vmatrix}$$



一、矢量代数

(ii) 矢量三重积

$$\vec{A}\times(\vec{B}\times\vec{C})=\vec{B}(\vec{A}\cdot\vec{C})-\vec{C}(\vec{A}\cdot\vec{B})$$ BAC-CAB规则

$$(\vec{A}\times\vec{B})\times\vec{C}=-\vec{C}\times(\vec{A}\times\vec{B})=-\vec{A}(\vec{B}\cdot\vec{C})+\vec{B}(\vec{A}\cdot\vec{C})$$

$$(\vec{A}\times\vec{B})\cdot(\vec{C}\times\vec{D})=[\vec{D}\times(\vec{A}\times\vec{B})]\cdot\vec{C}=[\vec{A}(\vec{D}\cdot\vec{B})-\vec{B}(\vec{D}\cdot\vec{A})]\cdot\vec{C}$$

$$=(\vec{A}\cdot\vec{C})(\vec{D}\cdot\vec{B})-(\vec{B}\cdot\vec{C})(\vec{D}\cdot\vec{A})$$

$$\vec{A}\times[\vec{B}\times(\vec{C}\times\vec{D})]=\vec{B}[\vec{A}\cdot(\vec{C}\times\vec{D})]-(\vec{C}\times\vec{D})(\vec{A}\cdot\vec{B})$$

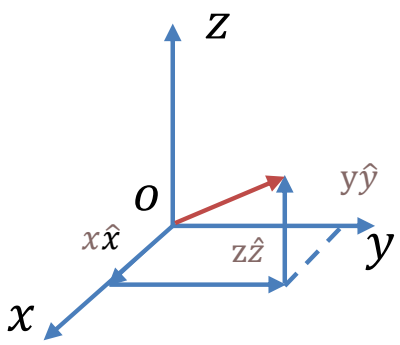
1.4 位置、位移及间隔矢量

在特定的坐标系，**位置矢量指该位置距离坐标原点的矢量**，在坐标基矢中展开，以直角坐标系为例

$$\vec{r} = x\hat{x} + y\hat{y} + z\hat{z}$$

$$|\vec{r}| = \sqrt{x^2 + y^2 + z^2}$$ 大小

$$\hat{r} = \frac{\vec{r}}{r} = \frac{x\hat{x} + y\hat{y} + z\hat{z}}{\sqrt{x^2 + y^2 + z^2}}$$ 单位矢量

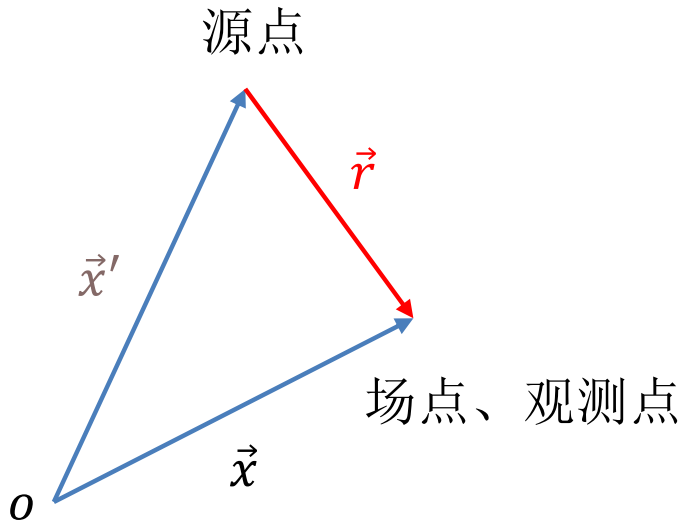


$$d\vec{l} = dx\hat{x} + dy\hat{y} + dz\hat{z}$$ 无限小位移

一、矢量代数

间隔矢量：场点到源点的矢量

在电动力学中，约定源点$\vec{x}$,场点为$\vec{x}'$



$$\vec{r} = \vec{x} - \vec{x}'$$

$$|\vec{r}| = |\vec{x} - \vec{x}'|$$

$$\hat{r} = \frac{\vec{r}}{|\vec{r}|} = \frac{\vec{x} - \vec{x}'}{|\vec{x} - \vec{x}'|}$$

直角坐标系中

$$\vec{r} = (x - x')\hat{x} + (y - y')\hat{y} + (z - z')\hat{z}$$

$$|\vec{r}| = \sqrt{(x - x')^2 + (y - y')^2 + (z - z')^2}$$

$$\hat{r} = \frac{\vec{r}}{r} = \frac{(x - x')\hat{x} + (y - y')\hat{y} + (z - z')\hat{z}}{\sqrt{(x - x')^2 + (y - y')^2 + (z - z')^2}}$$

二、微分运算

- 单变量函数$f(x)$微分的运算

$$df = (\frac{df}{dx})dx \qquad \frac{df}{dx}$$ 几何解释：斜率

- 三变量函数微分,例如温度$T(x,y,z)$的微分运算，由偏微分

$$dT = (\frac{\partial T}{\partial x})dx + (\frac{\partial T}{\partial y})dy + (\frac{\partial T}{\partial z})dz$$

$$= (\frac{\partial T}{\partial x}\hat{x} + \frac{\partial T}{\partial y}\hat{y} + \frac{\partial T}{\partial z}\hat{z}) \cdot d\hat{l}$$

$$= \nabla T \cdot d\hat{l}$$

∇ 算符（Nabla），矢量微分算符

$$\nabla = \hat{x}\frac{\partial}{\partial x} + \hat{y}\frac{\partial}{\partial y} + \hat{z}\frac{\partial}{\partial z}$$

∇T 几何解释：函数增加的最大的方向及最大的值，梯度

二、微分运算

∇算符运算 $\nabla = \hat{x}\frac{\partial}{\partial x} + \hat{y}\frac{\partial}{\partial y} + \hat{z}\frac{\partial}{\partial z}$

几何解释

- 梯度，作用在标量上∇T $\nabla T = \frac{\partial T}{\partial x}\hat{x} + \frac{\partial T}{\partial y}\hat{y} + \frac{\partial T}{\partial z}\hat{z}$
- 散度，作用在矢量上$\nabla \cdot \vec{v}$ $\nabla \cdot \vec{v} = \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z}$ 是否有源
- 旋度，作用在矢量上$\nabla \times \vec{v}$ $\nabla \times \vec{v} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ v_x & v_y & v_z \end{vmatrix}$ 是否有旋转

$$= \hat{x}\left(\frac{\partial v_z}{\partial y} - \frac{\partial v_y}{\partial z}\right) + \hat{y}\left(\frac{\partial v_x}{\partial z} - \frac{\partial v_z}{\partial x}\right) + \hat{z}\left(\frac{\partial v_y}{\partial x} - \frac{\partial v_z}{\partial y}\right)$$

二、微分运算

一般微分的运算

$$\frac{d}{dx}(f+g)=\frac{df}{dx}+\frac{dg}{dx}$$

$$\frac{d}{dx}(kf)=k\frac{df}{dx}(k,\text{常数})$$

$$\frac{d}{dx}(fg)=f\frac{dg}{dx}+g\frac{df}{dx}$$

$$\frac{d}{dx}\left(\frac{f}{g}\right)=\frac{g\frac{df}{dx}-f\frac{dg}{dx}}{g^2}$$

∇的运算

$$\nabla(f+g)=\nabla f+\nabla g$$

$$\nabla\cdot\left(\vec{A}+\vec{B}\right)=\nabla\cdot\vec{A}+\nabla\cdot\vec{B}$$

$$\nabla\times\left(\vec{A}+\vec{B}\right)=\nabla\times\vec{A}+\nabla\times\vec{B}$$

$$\nabla(kf)=k\nabla f$$

$$\nabla\cdot\left(k\vec{A}\right)=k\left(\nabla\cdot\vec{A}\right)$$

$$\nabla\times\left(\vec{k}\vec{A}\right)=\left(k\nabla\times\vec{A}\right)$$

二、微分运算

形成标量　　fg或$\vec{A}\cdot\vec{B}$

形成矢量　　$f\vec{A}$或$\vec{A}\times\vec{B}$

$$\nabla(fg)=f\nabla g+g\nabla f$$

$$\nabla\left(\vec{A}\cdot\vec{B}\right)=\vec{A}\times\left(\nabla\times\vec{B}\right)+\vec{B}\times\left(\nabla\times\vec{A}\right)+\left(\vec{A}\cdot\nabla\right)\vec{B}+\left(\vec{B}\cdot\nabla\right)\vec{A}$$

$$\nabla\cdot\left(f\vec{A}\right)=f\nabla\cdot\vec{A}+\vec{A}\cdot(\nabla f)$$

$$\nabla\cdot\left(\vec{A}\times\vec{B}\right)=\vec{B}\cdot\left(\nabla\times\vec{A}\right)-\vec{A}\cdot\left(\nabla\times\vec{B}\right)$$

二、微分运算

$$\nabla \times \left(f\vec{A}\right) = f\nabla \times \vec{A} + \vec{A} \times (\nabla f)$$

$$\nabla \times \left(\vec{A} \times \vec{B}\right) = \vec{A}\left(\nabla \cdot \vec{B}\right) - \vec{B}\left(\nabla \cdot \vec{A}\right) + \left(\vec{B} \cdot \nabla\right)\vec{A} - \left(\vec{A} \cdot \nabla\right)\vec{B}$$

$$\nabla \cdot \left(f\vec{A}\right) = \frac{\partial}{\partial x}(fA_x) + \frac{\partial}{\partial y}\left(fA_y\right) + \frac{\partial}{\partial z}(fA_z)$$

$$= \left(\frac{\partial f}{\partial x}A_x + f\frac{\partial A_x}{\partial x}\right) + \left(\frac{\partial f}{\partial y}A_y + f\frac{\partial A_y}{\partial y}\right) + \left(\frac{\partial f}{\partial z}A_z + f\frac{\partial A_z}{\partial z}\right)$$

$$= f\nabla \cdot \vec{A} + \vec{A} \cdot (\nabla f)$$

二、微分运算

二阶微分

$$\nabla\cdot(\nabla T)=\left(\frac{\partial}{\partial x}\hat{x}+\frac{\partial}{\partial y}\hat{y}+\frac{\partial}{\partial z}\hat{z}\right)\cdot\left(\frac{\partial T}{\partial x}\hat{x}+\frac{\partial T}{\partial y}\hat{y}+\frac{\partial T}{\partial z}\hat{z}\right)=\frac{\partial^2 T}{\partial x^2}+\frac{\partial^2 T}{\partial y^2}+\frac{\partial^2 T}{\partial z^2}=\boxed{\nabla^2 T}$$

$$\nabla\times(\nabla T)=0$$

$\boxed{\nabla(\nabla\cdot\vec{v})}$

$$\boxed{\nabla^2=\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}+\frac{\partial^2}{\partial z^2}}$$ 拉普拉斯算符

$$\nabla\cdot(\nabla\times\vec{v})=0$$

$$\nabla\times(\nabla\times\vec{v})=\nabla(\nabla\cdot\vec{v})-\nabla^2\vec{v}$$

注意：1. $\nabla(\nabla\cdot\vec{v})$与$\nabla^2\vec{v}$是不同的，前者是散度的梯度，后者是拉普拉斯算符

2. 只存在散度的梯度以及拉普拉斯运算两种

二、微分运算

$$\nabla \times (\nabla T) = 0$$

证明：

$$\nabla \times (\nabla T) = \hat{x}\left[\frac{\partial}{\partial y}\left(\frac{\partial T}{\partial z}\right) - \frac{\partial}{\partial z}\left(\frac{\partial T}{\partial y}\right)\right] + \hat{y}\left[\frac{\partial}{\partial z}\left(\frac{\partial T}{\partial x}\right) - \frac{\partial}{\partial x}\left(\frac{\partial T}{\partial z}\right)\right] + \hat{z}\left[\frac{\partial}{\partial x}\left(\frac{\partial T}{\partial y}\right) - \frac{\partial}{\partial y}\left(\frac{\partial T}{\partial x}\right)\right]$$

$$= 0$$

$$\frac{\partial}{\partial y}\left(\frac{\partial T}{\partial z}\right) = \frac{\partial}{\partial z}\left(\frac{\partial T}{\partial y}\right)$$

$$\nabla \cdot (\nabla \times \vec{v}) = 0$$

- 梯度场无旋
- 旋度场无源

三、积分运算

线积分

$$\int_S \vec{v} \cdot d\vec{l}$$

存在某些特殊矢量（$\nabla \times \vec{v} = 0$），积分与路径无关

面积分

$$\int_S \vec{v} \cdot d\vec{a}$$

存在某些特殊矢量（$\nabla \cdot \vec{v} = 0$），积分与面无关，仅取决于边界

体积分

$$\int_v \vec{v} dV = \int_v \left(v_x \hat{x} + v_y \hat{x} + v_z \hat{x}\right) dV = \hat{x} \int_v v_x dz + \hat{y} \int_v v_y dz + \hat{z} \int_v v_z dz$$

三、积分运算

积分的基本理论

$$\int_a^b \frac{df}{dx}dx = f(b) - f(a)$$

$$dT = (\nabla T)\cdot d\vec{l}_1 \qquad \int_a^b (\nabla T)\cdot d\vec{l} = T(\vec{b}) - T(\vec{a})$$

$$\int (\nabla\cdot\vec{v})d\tau = \oint_s \vec{v}\cdot d\vec{a} \qquad \text{高斯公式}$$

$$\int_s (\nabla\times\vec{v})\cdot d\vec{a} = \oint_p \vec{v}\cdot d\vec{l} \qquad \text{斯托克斯公式}$$

三、积分运算

分部积分

$$\frac{d}{dx}(fg) = f\frac{dg}{dx} + g\frac{df}{dx}$$

一维到三维

$$\int_a^b \frac{d}{dx}(fg)dx = fg\Big|_a^b = \int_a^b f\left(\frac{dg}{dx}\right)dx + \int_a^b g\left(\frac{df}{dx}\right)dx$$

$$\int_a^b f\left(\frac{dg}{dx}\right)dx = -\int_a^b g\left(\frac{df}{dx}\right)dx + \boxed{fg\Big|_a^b}$$

边界项

$$\nabla\cdot\left(f\vec{A}\right) = f\left(\nabla\cdot\vec{A}\right) + \vec{A}\cdot(\nabla f)$$

$$\int \nabla\cdot\left(f\vec{A}\right)d\mathrm{V} = \int f\left(\nabla\cdot\vec{A}\right)d\mathrm{V} + \int \vec{A}\cdot(\nabla f)dV = \oint f\vec{A}\cdot d\vec{a}$$

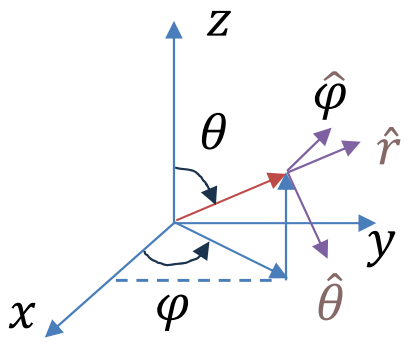
$$\int_v f(\nabla\cdot\vec{A})d\mathrm{V} = -\int_v \vec{A}\cdot(\nabla f)dV + \boxed{\oint_S f\vec{A}\cdot d\vec{a}}$$

边界项

四、曲线坐标系

4.1 球坐标

$$\begin{cases} x = r\sin\theta\cos\varphi \\ y = r\sin\theta\sin\varphi \\ z = r\cos\theta \end{cases}$$



$$\vec{A} = A_r\hat{r} + A_\theta\hat{\theta} + A_\varphi\hat{\varphi} \qquad \hat{r}\times\theta = \hat{\varphi}$$

取值范围：

$r: 0 \to \infty$

极角$\theta: (0, \pi)$

方位角φ：$(0, 2\pi)$

$$\begin{cases} \hat{r} = \sin\theta\cos\varphi\hat{x} + \sin\theta\sin\varphi\hat{y} + \cos\theta\hat{z} \\ \hat{\theta} = \cos\theta\cos\varphi\hat{x} + \cos\theta\sin\varphi\hat{y} - \sin\theta\hat{z} \\ \hat{\varphi} = -\sin\theta\hat{x} + \cos\varphi\hat{y} \end{cases}$$

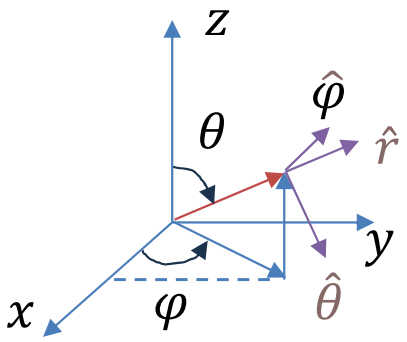
$\hat{r}, \hat{\theta}, \hat{\varphi}$随位置的改变而改变

$$\hat{z} = \cos\theta\hat{r} - \sin\theta\hat{\theta}$$

四、曲线坐标系

- 线元

$$\begin{cases} dl_r = dr \\ dl_\theta = rd\theta \\ dl_\varphi = rsin\theta d\varphi \end{cases}$$



- 体积元

$$dl_r dl_\theta dl_\varphi = r^2 dr d\theta sin\theta d\varphi$$

- 面积元，依赖于所选取的面

✓ 若选取球面，r为常数，$d\vec{s} = dl_\theta\, dl_\varphi \hat{r} = r^2 sin\theta d\theta d\varphi \hat{r}$

✓ 若选取x-y平面，θ为常数$(\theta = \pi/2)$，$d\vec{s} = dl_\varphi\, dl_r \hat{r} = r dr d\varphi \hat{\theta}$

四、曲线坐标系

- 用T表示任意标量场，用$\vec{v}$表示任意矢量场，后续在介绍柱坐标的时候用相同的约定

$$\frac{\partial T}{\partial x}=\frac{\partial T}{\partial r}\left(\frac{\partial r}{\partial x}\right)+\frac{\partial T}{\partial \theta}\left(\frac{\partial \theta}{\partial x}\right)+\frac{\partial T}{\partial \varphi}\left(\frac{\partial \varphi}{\partial x}\right)$$，其他变量做相同的代换

$$\nabla T=\frac{\partial T}{\partial x}\hat{x}+\frac{\partial T}{\partial y}\hat{y}+\frac{\partial T}{\partial z}\hat{z}$$

- 梯度

$$\nabla T=\frac{\partial T}{\partial r}\hat{r}+\frac{\partial T}{r\partial \theta}\hat{\theta}+\frac{1}{r\sin\theta}\frac{\partial T}{\partial \varphi}\hat{\varphi}$$

四、曲线坐标系

- 散度

$$\nabla \cdot \vec{v} = \frac{1}{r^2}\frac{\partial}{\partial r}(r^2 v_r) + \frac{1}{r\sin\theta}\frac{\partial}{\partial \theta}(\sin\theta\, v_\theta) + \frac{1}{r\sin\theta}\frac{\partial v_\varphi}{\partial \varphi}$$

- 旋度

$$\nabla \times \vec{v} = \frac{1}{r\sin\theta}\left[\frac{\partial}{\partial \theta}(\sin\theta\, v_\varphi) - \frac{\partial v_\theta}{\partial \varphi}\right]\hat{r} + \frac{1}{r}\left[\frac{1}{\sin\theta}\frac{\partial v_r}{\partial \varphi} - \frac{\partial}{\partial r}(r v_\varphi)\right]\hat{\theta} + \frac{1}{r}\left[\frac{\partial}{\partial r}(r v_\theta) - \frac{\partial v_r}{\partial \theta}\right]\hat{\varphi}$$

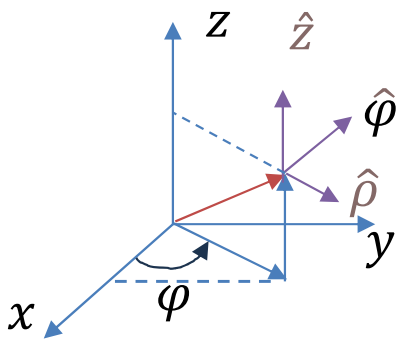
- 标量二阶导

$$\nabla^2 T = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial T}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial \theta}\left(\sin\theta\frac{\partial T}{\partial \theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2}{\partial \varphi^2}$$

四、曲线坐标系

4.2 柱坐标

$$\begin{cases} x = \rho cos\varphi \\ y = \rho sin\varphi \\ z = z \end{cases}$$



$$\vec{A} = A_\rho \hat{\rho} + A_\varphi \hat{\varphi} + A_z \hat{z} \qquad \hat{\rho} \times \hat{\varphi} = \hat{z}$$

$$\begin{cases} \hat{\rho} = cos\varphi \hat{x} + sin\varphi \hat{y} \\ \hat{\varphi} = -sin\varphi \hat{x} + cos\varphi \hat{y} \\ \hat{z} = \hat{z} \end{cases}$$

$\hat{\rho}, \hat{\varphi}, \hat{z}$随位置的改变而改变

四、曲线坐标系

- 线元

$$\begin{cases} dl_\rho = d\rho \\ dl_\varphi = rd\varphi \\ dl_z = dz \end{cases}$$

- 体积元

$$dl_\rho dl_\theta dl_z = \rho d\rho d\varphi dz$$

- 取值范围

$$\rho: \ 0 \to \infty$$
$$\varphi: \ 0 \to 2\pi$$
$$z: \ -\infty \to \infty$$

四、曲线坐标系

- 梯度

$$\nabla T = \frac{\partial T}{\partial \rho}\hat{\rho} + \frac{1}{\rho}\frac{\partial T}{\partial \varphi}\hat{\varphi} + \frac{\partial T}{\partial z}\hat{z}$$

- 散度

$$\nabla \cdot \vec{v} = \frac{1}{\rho}\frac{\partial}{\partial \rho}(\rho v_\rho) + \frac{1}{\rho}\frac{\partial v_\varphi}{\partial \varphi} + \frac{\partial v_z}{\partial z}$$

- 旋度

$$\nabla \times \vec{v} = \left(\frac{1}{\rho}\frac{\partial v_z}{\partial \varphi} - \frac{\partial v_\varphi}{\partial z}\right)\hat{\rho} + \left(\frac{\partial v_\rho}{\partial z} - \frac{\partial v_z}{\partial \rho}\right)\hat{\varphi} + \frac{1}{\rho}\left(\frac{\partial}{\partial \rho}(\rho v_\varphi) - \frac{\partial v_\rho}{\partial \varphi}\right)\hat{z}$$

- 标量二阶导

$$\nabla^2 T = \frac{1}{\rho}\frac{\partial}{\partial \rho}\left(\rho\frac{\partial T}{\partial \rho}\right) + \frac{1}{\rho^2}\frac{\partial^2 T}{\partial \varphi^2} + \frac{\partial^2 T}{\partial z^2}$$

五、狄拉克 δ 函数

5.1 对于$\frac{\hat{r}}{r^2}$的散度

$$\nabla \cdot \vec{v} = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 v_r\right) + \frac{1}{r sin\theta}\frac{\partial}{\partial \theta}\left(sin\theta v_\theta\right) + \frac{1}{r sin\theta}\frac{\partial v_\varphi}{\partial \varphi}$$

$$\vec{v} = \frac{\hat{r}}{r^2}$$

$$\nabla \cdot \vec{v} = \frac{1}{r^2}\frac{\partial}{\partial r}(r^2 v_r) = 0$$

$$\oint \vec{v} \cdot d\vec{S} = \oint \left(\frac{1}{r^2}\hat{r}\right) \cdot r^2 sin\theta d\theta d\varphi \hat{r} = \int_0^\pi sin\theta d\theta \int_0^{2\pi} d\varphi = 4\pi$$

高斯公式

$$\oint \vec{v} \cdot d\vec{S} = \int_V \nabla \cdot \vec{v}\, dV$$

因此r=0时$\nabla \cdot \vec{v}$不为0，而是$\nabla \cdot \vec{v} = 4\pi\delta(\vec{r})$,δ函数的归一性可以保证高斯公式成立

五、狄拉克 δ 函数

5.2 一维狄拉克δ函数(广义函数)

$$\delta(x)=\begin{cases}\infty & \text{if } x=0\\ 0 & \text{if } x\neq 0\end{cases}$$

$$\int_{-\infty}^{\infty}\delta(x)dx=1$$

对于函数$f(x)$

$$\int_{-\infty}^{\infty}f(x)\delta(x)dx=\int_{-\infty}^{\infty}f(0)\delta(x)dx=f(0)\int_{-\infty}^{\infty}\delta(x)\,dx=f(0)$$

五、狄拉克 δ 函数

狄拉克δ函数的挑选性

$$\delta(x-a)=\begin{cases}\infty & \text{if } x=a\\ 0 & \text{if } x\neq a\end{cases}$$

$$\int_{-\infty}^{\infty}\delta(x-a)dx=1$$

对于函数$f(x)$

$$\int_{-\infty}^{\infty}f(x)\delta(x-a)dx=\int_{-\infty}^{\infty}f(a)\delta(x-a)dx$$
$$=f(a)\int_{-\infty}^{\infty}\delta(x-a)\,dx=f(a)$$

五、狄拉克 δ 函数

狄拉克**δ**函数其他性质

$\delta(kx)=\frac{1}{|k|}\delta(x)$，以下给出证明

计算$\int_{-\infty}^{\infty}f(x)\delta(kx)dx$

令$\mathrm{y}=kx,\ x=\frac{\mathrm{y}}{k}\ ,\ dx=\frac{\mathrm{dy}}{k}$

若$k>0,$

$$\int_{-\infty}^{\infty}f(x)\delta(kx)dx=\int_{-\infty}^{\infty}f\left(\frac{\mathrm{y}}{k}\right)\frac{1}{k}\delta(y)dy=\frac{1}{k}\int_{-\infty}^{\infty}f\left(\frac{\mathrm{y}}{k}\right)\delta(y)dy=\frac{1}{k}f(0)$$

若$k<0,$

$$\int_{-\infty}^{\infty}f(x)\delta(kx)dx=-\int_{-\infty}^{\infty}f\left(\frac{\mathrm{y}}{k}\right)\frac{1}{k}\delta(y)dy=-\frac{1}{k}\int_{-\infty}^{\infty}f\left(\frac{\mathrm{y}}{k}\right)\delta(y)dy=-\frac{1}{k}f(0)$$

五、狄拉克 δ 函数

狄拉克δ函数其他性质

因此

$$\int_{-\infty}^{\infty} f(x)\delta(kx)dx = \frac{1}{|k|}\int_{-\infty}^{\infty} f(x)\delta(x)dx$$

即

$$\delta(kx) = \frac{1}{|k|}\delta(x)$$

得证

五、狄拉克 δ 函数

三维狄拉克**δ**函数**(**广义函数**)**

$$\delta(\vec{r}) = \delta(x)\delta(y)\delta(z)$$

其中$\vec{r}= x\hat{x} + y\hat{y}+ z\hat{z}$

对全空间积分

$$\int \delta(\vec{r})\mathrm{dV} = \int_{-\infty}^{\infty} \delta(x)dx \int_{-\infty}^{\infty} \delta(y)\,dy \int_{-\infty}^{\infty} \delta(z)dz = 1$$

$$\int f\,(\vec{r})\delta(\vec{r} - \vec{a})d\mathrm{V} = f(\vec{a})$$

五、狄拉克 δ 函数

三维狄拉克**δ**函数(广义函数)

$$\nabla \cdot \frac{\hat{\mathrm{r}}}{r^2} = 4\pi\delta(\vec{r})$$

$$\nabla \cdot \frac{\hat{\imath}}{\imath} = 4\pi\delta(\vec{\imath})$$

其中$\vec{\imath} = \vec{r} - \vec{r}'$

$$\nabla\left(\frac{1}{\imath}\right) = -\frac{\hat{\imath}}{\imath^2}$$

$$\nabla \cdot \nabla\left(\frac{1}{\imath}\right) = \nabla^2\left(\frac{1}{\imath}\right) = -4\pi\delta(\vec{\imath})$$

狄拉克 δ 函数

1. 证明$x\frac{d}{dx}\big(\delta(x)\big)=-\delta(x)$

2. 阶梯函数$\theta(x)=\begin{cases}1, & x>0\\ 0, & x\le 0\end{cases}$，证明$\frac{d\theta}{dx}=\delta(x)$

第一周作业：P31，1.1，1.2，1.3, 1.4